

EXPERIMENT 10

Email Header Spoofing Analysis

AIM

Parse simulated email headers and compare From, Reply-To, Return-Path, Received and authentication results to identify suspicious indicators.

ALGORITHM

1. Start.
2. Read/accept the required evidence data.
3. Process and analyze the data according to the experiment.
4. Apply the required threshold/rules/comparison.
5. Display the findings.
6. Stop.

CODE

```
import re
header="""From: attacker@example.com
Reply-To: support@different-domain.com
Return-Path: attacker@example.com
Authentication-Results: spf=fail; dkim=none"""
def get(f):
    m=re.search(rf"^{f}:\s*(.+)$",header,re.I|re.M)
    return m.group(1) if m else "Not available"
for f in ["From","Reply-To","Return-Path","Authentication-Results"]: print(f+":",get(f))
print("Indicator: Reply-To domain differs from From domain")
print("Indicator: SPF failed")
```

OUTPUT



```
Python 3.10.4 Shell
C:\Users\saish> cd C:\Users\saish\OneDrive\Desktop> dir
.
EXP-10.py
get

C:\Users\saish\OneDrive\Desktop> cd C:\Users\saish\OneDrive\Desktop> python3 EXP-10.py
From: attacker@example.com
Reply-To: support@different-domain.com
Return-Path: attacker@example.com
Authentication-Results: spf=fail; dkim=none
Indicator: Reply-To domain differs from From domain
Indicator: SPF failed
```

RESULT

The parser extracted the requested header fields and reported inconsistencies as suspicious indicators.